



Hopper Utilization Stats

2023 - 2024

Introduction

- Utilization stats for Hopper for the academic year running between July 2023 – June 2024 are presented.
- We start with a summary of all the jobs run then give a breakdown based on the Schools and Departments, looking at CPU/GPU utilization.
- We also look at the number of unique active users every month and top users based on the number of jobs submitted.
- In the second part, a summary of class accounts is given.

General Utilization Stats

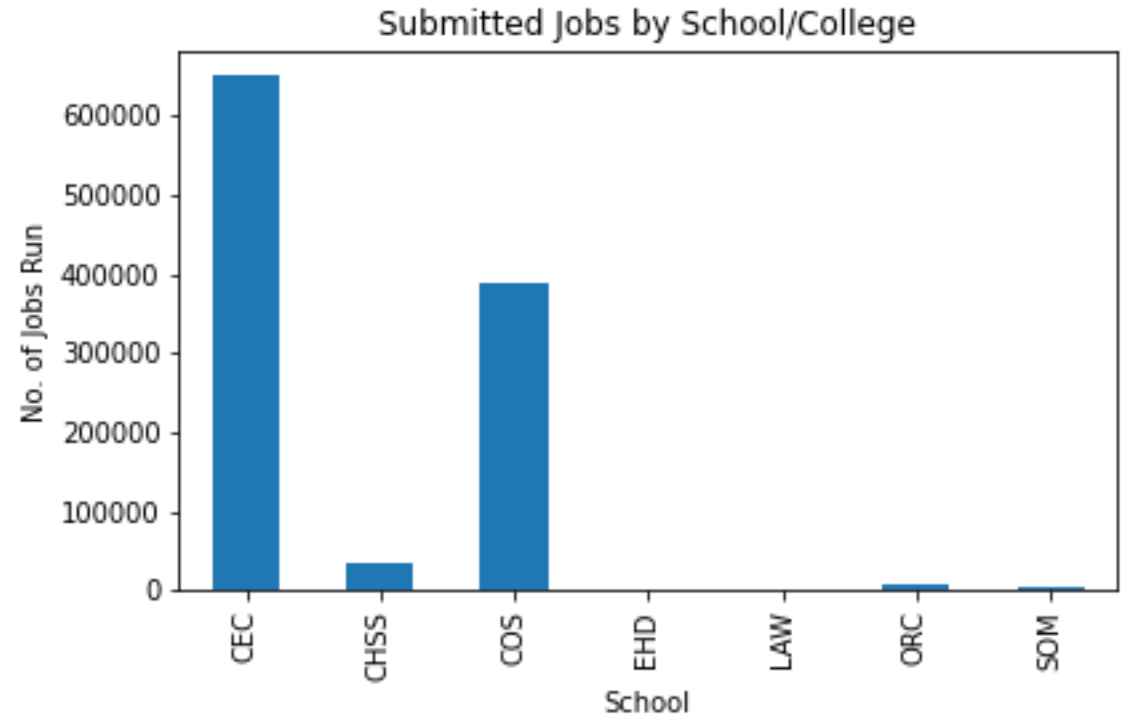
Submitted Jobs by the Different Schools

Of the departments that use the cluster, the College of Engineering and Computing has submitted the most jobs between July 2023 and June 2024, submitting 650262 jobs overall

The College of Education and Human Development has submitted the least, at a mere 165 jobs submitted.

CPU Jobs	GPU Jobs	orc-test
971496	114342	1114

College	Submit
CEC	650262
CHSS	36435
COS	389958
EHD	165
LAW	337
ORC	6771
SOM	3024
Total	1086952



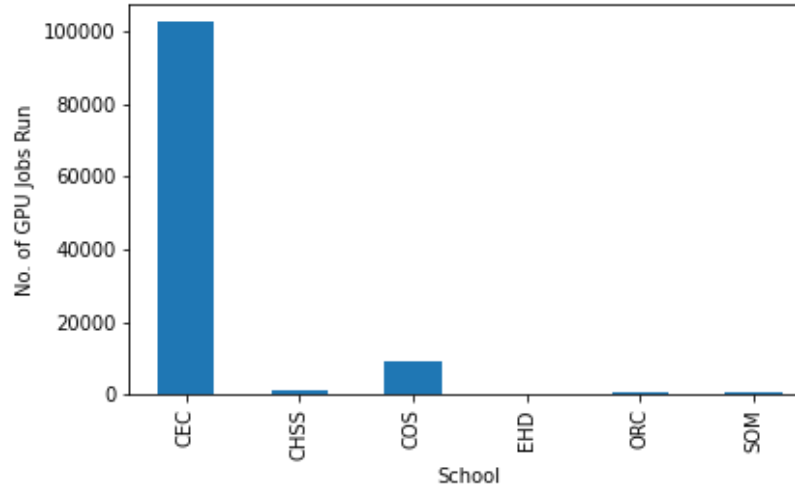
Summary of all GPU Jobs

GPU usage is highest by the College of Engineering and Computing by a large margin

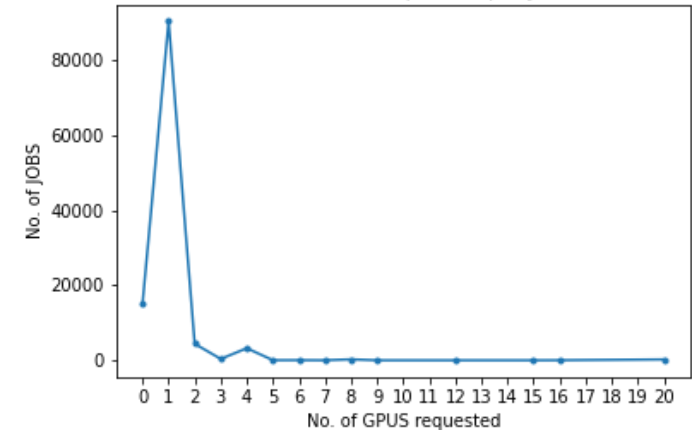
The majority of GPU jobs request only one GPU

The most commonly requested type of GPU is the 80gb GPU

Submitted GPU Jobs by School

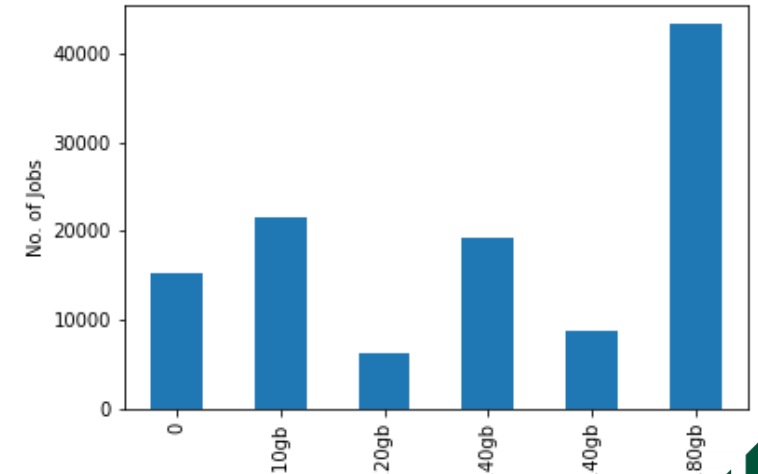


No. of GPUs Requested per Job



College	Submit
CEC	102510
CHSS	1111
COS	9279
EHD	22
ORC	452
SOM	968
114342	

Types of GPU Requested

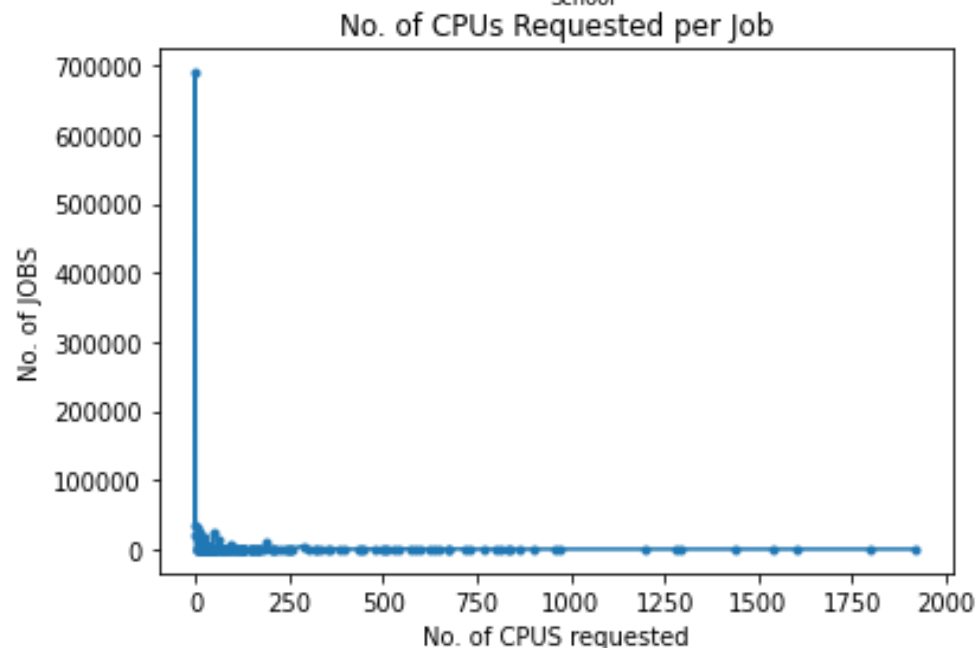
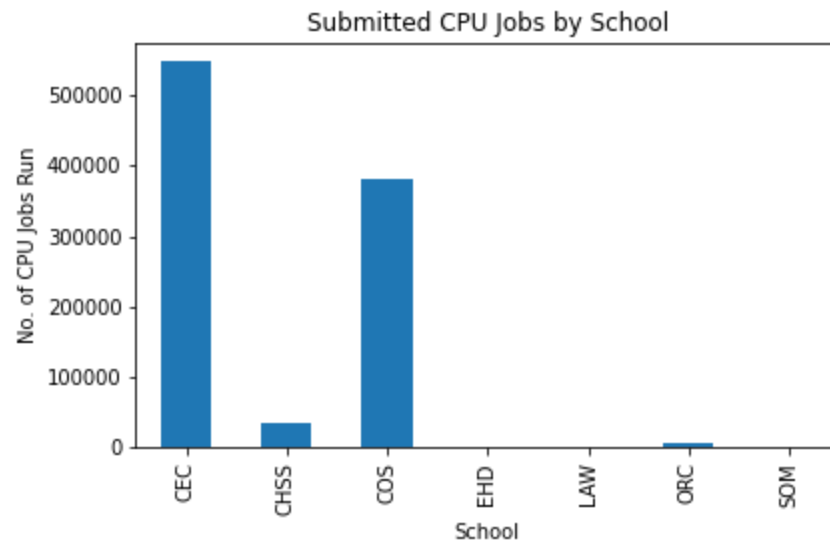
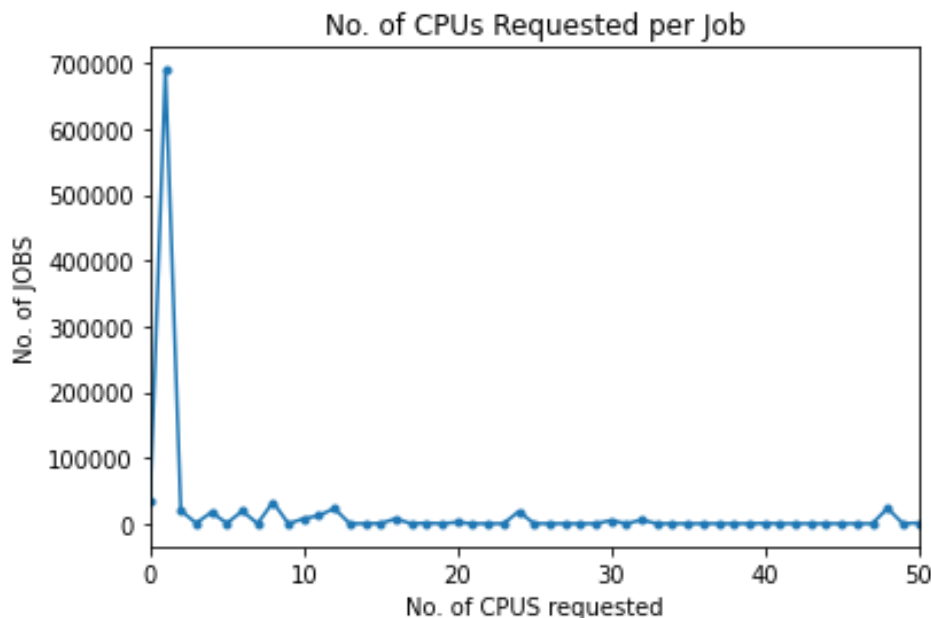


Summary of all CPU Jobs

CEC and COS remain the most common requestors for CPU jobs.

The vast majority of CPU jobs request less than 10 CPUs

College	Submit
CEC	547752
CHSS	35324
COS	380679
EHD	143
LAW	337
ORC	5205
SOM	2056
	971496

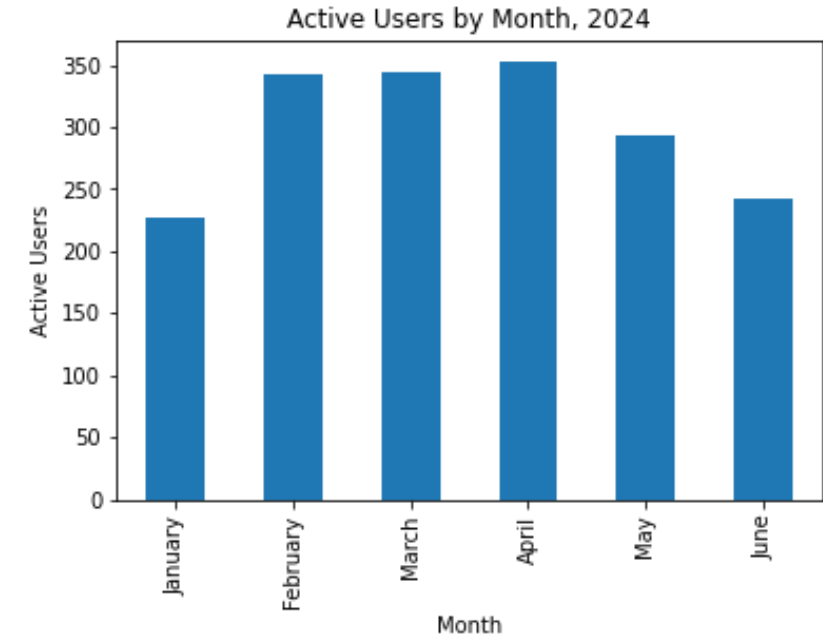
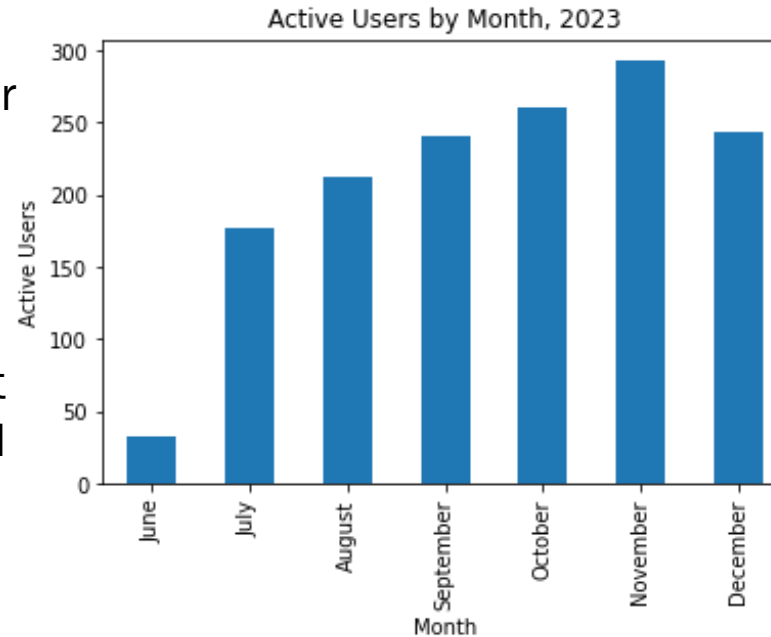


Unique Users per Month on Hopper

In the Fall Semester of 2023, Cluster Usage peaks in November towards the term.

In the Spring Semester of 2024, Cluster usage instead peaks towards the middle of the term. It also starts at a much higher level than in the previous semester

Year	Month	Active Users
2023	June	32
2023	July	177
2023	August	212
2023	September	240
2023	October	260
2023	November	293
2023	December	243
2024	January	227
2024	February	342
2024	March	344
2024	April	352
2024	May	294
2024	June	243



Top users submitting jobs

User	Dept	Department	College	Submits
awelbaum	stat	Department of Statistics	CEC	206152
tzhang23	ceie	Department of Civil, Environmental, and Infrastructure Engineering	CEC	87002
ohussei	phys	Department of Physics and Astronomy	COS	67008
sshakya4	biol	Department of Biology	COS	65492
ylu22	cs	Department of Computer Science	CEC	39370
jmckaig	phys	Department of Physics and Astronomy	COS	38987
wdai	stat	Department of Statistics	CEC	31025
jshao4	stat	Department of Statistics	CEC	30581
bthapa3	phys	Department of Physics and Astronomy	COS	30379
bcash	aoes	Department of Atmospheric, Oceanic and Earth Sciences	COS	29077
gmarkel	econ	Department of Economics	CHSS	28686
syarikar	cs	Department of Computer Science	CEC	26253
ddas6	cs	Department of Computer Science	CEC	25858
mherceg	cs	Department of Computer Science	CEC	25549
zwang48	ece	Department of Electrical and Computer Engineering	CEC	22011
schang31	aoes	Department of Atmospheric, Oceanic and Earth Sciences	COS	18836
ikhatri	phys	Department of Physics and Astronomy	COS	17378
jsheballi	phys	Department of Physics and Astronomy	COS	15123
rmacon2	stat	Department of Statistics	CEC	13306
jamankw	chem	Department of Chemistry and Biochemistry	COS	11537

GPU Utilization by Department

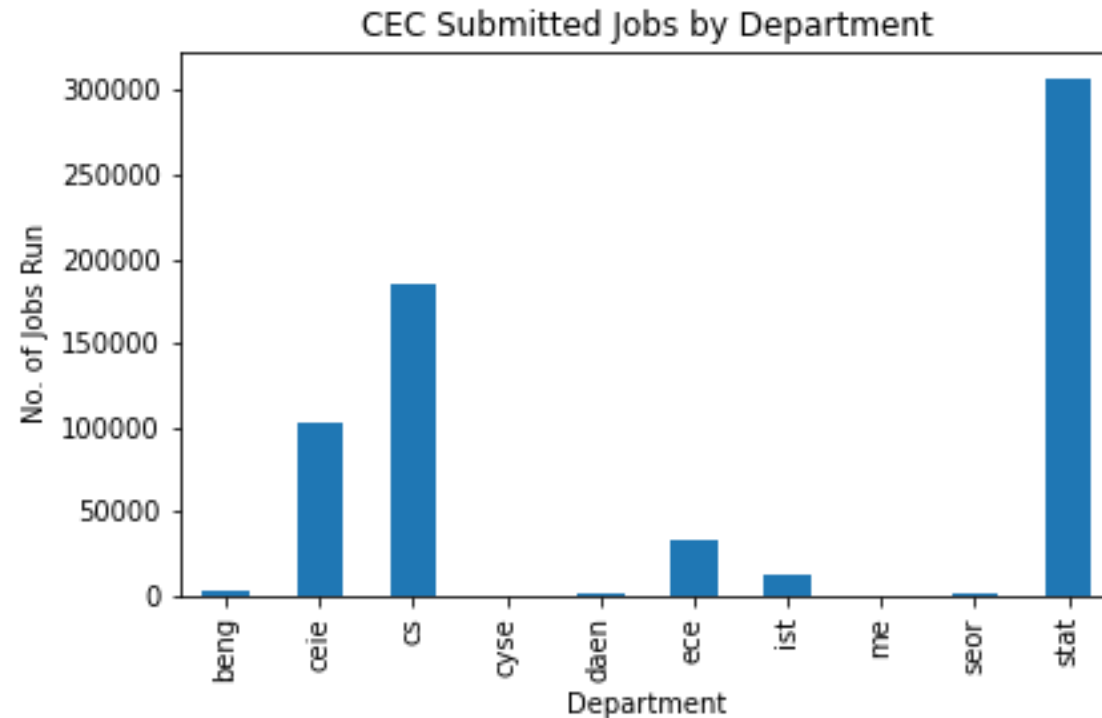
Department	Submit	RunTime (Hrs)	NCPUS	NGPUS	MEM (GB)
Department of Computer Science	55601	276204.92	352466	62363	5651527
Department of Electrical and Computer Engineering	30040	79567.39	90790	26704	1610597
Department of Computational and Data Sciences	3749	76056.45	35988	3302	83665
Department of Information Sciences and Technology	10667	65480.59	57338	12245	631755
School of Business	968	30963.98	8012	1171	101433
Department of Atmospheric, Oceanic and Earth Sciences	2282	16729.52	21356	2801	565710
Department of Physics and Astronomy	1182	11520.38	11648	1094	51606
Department of Geography and Geoinformation Science	1194	8668.10	6620	1168	115584
Department of Systems Engineering and Operations Research	816	7381.58	4189	642	35024
Department of Chemistry and Biochemistry	435	7352.60	2810	1297	12196
Data Analytics and Engineering	1358	4601.74	17670	1457	102466
School of Systems Biology	86	4120.29	1933	162	3857
Department of Economics	278	3568.25	2197	484	17094
Department of Bioengineering	3445	3185.97	3379	3358	190889
Department of Psychology	515	2234.60	13412	503	45951
Department of English	318	2003.19	2466	329	24644
Sid and Reva Dewberry Department of Civil, Environmental, and Infrastructure Engineering	95	1268.11	837	85	4217
Department of Mathematical Sciences	345	542.03	3107	384	27218
Office of Research Computing	452	462.02	2897	562	11516
Department of Mechanical Engineering	344	415.50	10070	628	72204
Department of Statistics	97	358.66	384	109	1819
School of Education	22	264.07	44	22	176
Department of Cyber Security Engineering	47	32.16	951	32	7498
Department of Biology	6	0.07	384	9	600

CEC Jobs Breakdown

The College of Engineering and Computational Sciences is the most prominent user of cluster resources.

Interestingly, it is not the Computer Science department that makes the most usage of the cluster, but the Stats department.

Dept	Submit
beng	4004
ceie	103678
cs	184432
cyse	78
daen	1763
ece	34121
ist	12980
me	409
seor	1041
stat	307756
Total	650262

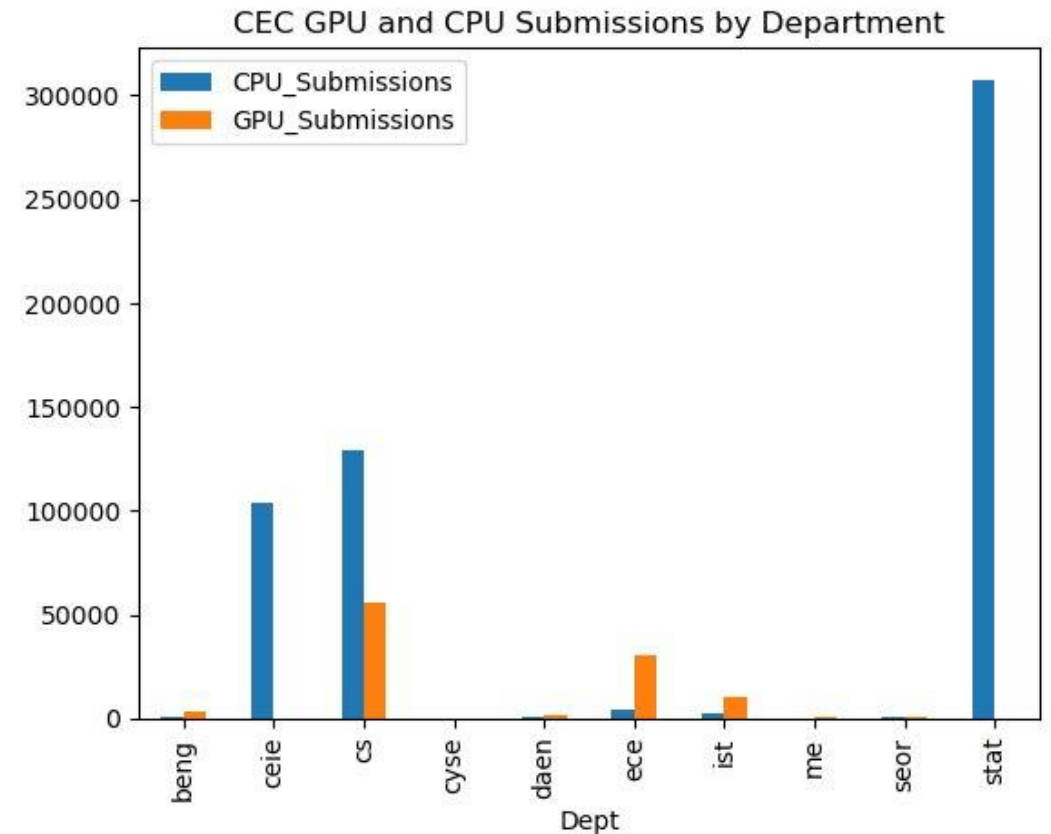


CEC Top User Board

User	Descr	Dept	College	Submit
rpatel17	carlotta domeniconi:cdomenic@gmu.edu	cs	CEC	7953
syarikar	hakan aydin:aydin@gmu.edu	cs	CEC	6153
ywang68	lily wang:lwang41@gmu.edu	stat	CEC	4962
yjiang25	kai zeng:kzeng2@gmu.edu	ece	CEC	3903
jkopsick	giorgio ascoli:ascoli@gmu.edu	beng	CEC	3493
tzhang23	lucas henneman:lhennem@gmu.edu	ceie	CEC	1898
hwu27	keren zhou:kzhou6@gmu.edu	cs	CEC	1878
akumardi	kevin p moran:kpmoran@gmu.edu	cs	CEC	1647
zwang48	wenwen jiang:wjiang8@gmu.edu	ece	CEC	1493
awelbaum	wanli qiao:wqiao@gmu.edu	stat	CEC	1398
yzhou31	ziwei zhu:zzhu20@gmu.edu	cs	CEC	901
kteimour	erion plaku:plaku@gmu.edu	cs	CEC	755
ylu22	erion plaku:plaku@gmu.edu	cs	CEC	562
mrasel	lucas henneman:lhennem@gmu.edu	ceie	CEC	511
rmacdon2	ben seiyon lee:slee287@gmu.edu	stat	CEC	484
xshan2	lucas henneman:lhennem@gmu.edu	ceie	CEC	425
ddas6	xuesu xiao:xiao@gmu.edu	cs	CEC	314
lhennem	lucas henneman:lhennem@gmu.edu	ceie	CEC	285
wdai	jiayang sun:jsun21@gmu.edu	stat	CEC	245
zhuang6	wanli qiao:wqiao@gmu.edu	stat	CEC	219
dkepplin	david kepplinger:dkepplin@gmu.edu	stat	CEC	218
mherceg	amarda shehu:ashehu@gmu.edu	cs	CEC	170
swei3	david kepplinger:dkepplin@gmu.edu	stat	CEC	132
jshao4	wanli qiao:wqiao@gmu.edu	stat	CEC	101
pbaldon	erion plaku:plaku@gmu.edu	cs	CEC	90
wqiao	wanli qiao:wqiao@gmu.edu	stat	CEC	40

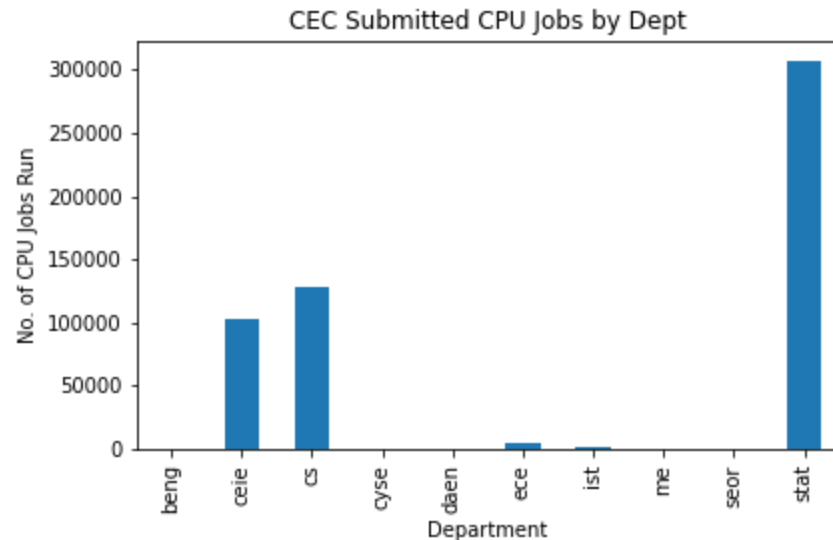
CEC CPU vs GPU Usage

- Overall, CPU jobs are more requested by the CEC than GPU jobs.
- Interestingly, the most prolific user of GPUs is not the Stats department (which has the highest overall usage of cluster resources among departments) but the CS department. Stats actually requests a very low number of GPU jobs.

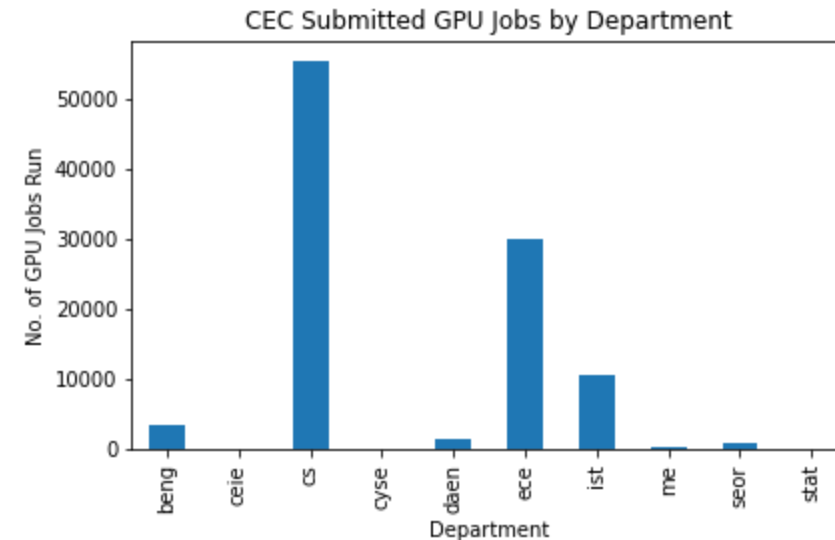


CEC CPU/GPU Tables

Dept	No. of CPU Jobs Submitted	CPU Hrs	MEM(GB)
beng	559	38952.36	31192.00
ceie	103583	2533548.95	2743116.94
cs	128831	945335.84	1405161.37
cyse	31	11555.15	7025.00
daen	405	23412.45	22835.00
ece	4081	11974263.06	829316.42
ist	2313	109349.78	95093.00
me	65	8299.56	7072.00
seor	225	134139.89	4110.00
stat	307659	4487273.25	6259955.80



Dept	No. of GPU Jobs Submitted	CPU Hrs	GPU Hrs	MEM(GB)
beng	3445	3185.97	3304.51	190889.00
ceie	95	2079.57	18004.60	4217.67
cs	55601	467928.81	4032679.38	5651527.65
cyse	47	32.16	858.65	7498.00
daen	1358	5796.67	77086.10	102466.43
ece	30040	87745.32	459433.28	1610597.67
ist	10667	103376.83	527849.24	631755.14
me	344	824.64	13388.19	72204.00
seor	816	8535.69	65827.71	35024.00
stat	97	377.59	1367.38	1819.00



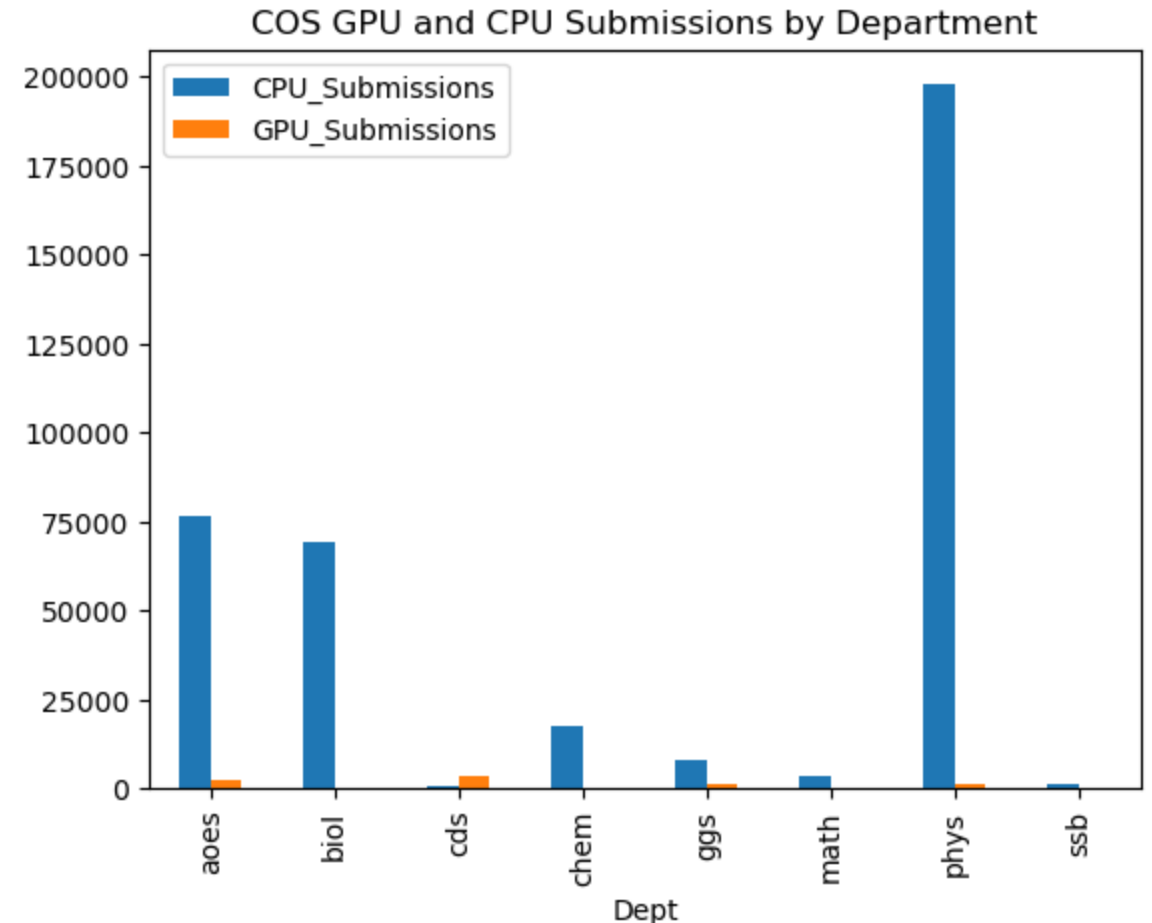
COS Jobs Breakdown

COS top user board

User	Descr	Dept	College	Submit
sshakya4	haw lim:hlim22@gmu.edu	biol	COS	38257
schang31	daniel tong:qtong@gmu.edu	aoes	COS	18563
jsheball	john sheball:jsheball	phys	COS	8607
jxing3	daniel tong:qtong@gmu.edu	aoes	COS	8435
bcash	benjamin a cash:bcash@gmu.edu	aoes	COS	7161
ohussei	yuri mishin:ymishin@gmu.edu	phys	COS	5126
ikhatri	yuri mishin:ymishin@gmu.edu	phys	COS	4399
yli74	daniel tong:qtong@gmu.edu	aoes	COS	3455
rkoju	yuri mishin:ymishin@gmu.edu	phys	COS	3029
yli84	yuri mishin:ymishin@gmu.edu	phys	COS	2751
nking21	predrag nikolic:pnikolic@gmu.edu	phys	COS	2376
bthapa3	igor mazin:imazin2@gmu.edu	phys	COS	2189
jmckaig	shobita satyapal:ssatyapa@gmu.edu	phys	COS	1442
yabere	estela blaisten:blaisten@gmu.edu	cds	COS	976
wmatzko	shobita satyapal:ssatyapa@gmu.edu	phys	COS	842
jamankw	robin couch:rcouch@gmu.edu	chem	COS	609
fkrueger	frank krueger:fkrueger@gmu.edu	ipn	COS	500
gschwert	igor mazin:imazin2@gmu.edu	phys	COS	275
ychiari	ylenia chiari:ychiari@gmu.edu	biol	COS	230
bdavis39	paul a dirmeyer:pdirmeye@gmu.edu	aoes	COS	158
nbearson	nick bearson:nbearson@gmu.edu	ggs	COS	73

COS CPU vs GPU Usage

- As with CEC, the COS exhibits more CPU usage than GPU usage
- The Physics Department makes the most use of cluster CPUs by a large margin among the COS departments

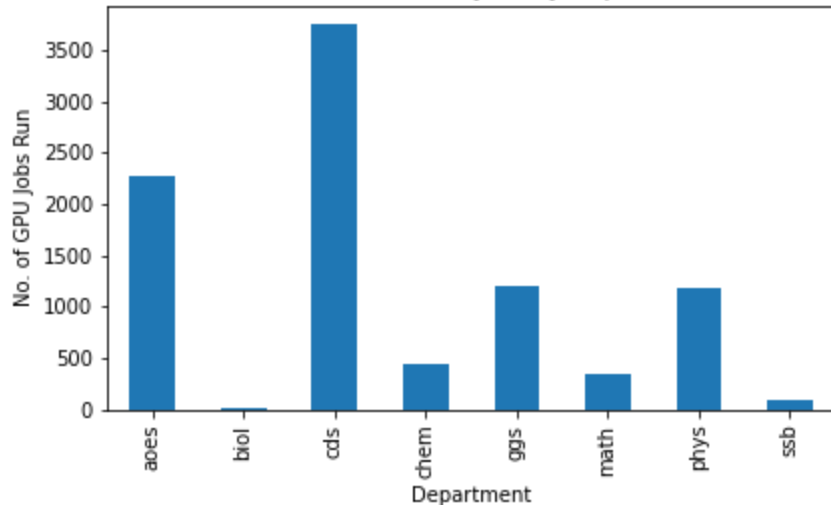


COS CPU/GPU Tables

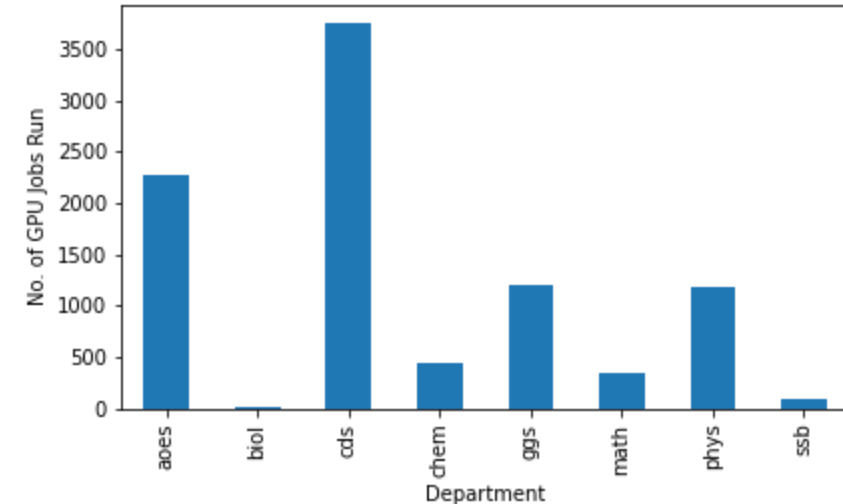
Dept	No. of CPU Jobs Submitted	CPU Hrs	MEM(GB)
aoes	76844	4826522.32	17944704.05
biol	69068	118100.97	1257552.25
cds	970	126423.92	61409.29
chem	17403	933354.94	1464413.12
esp	626	19642.86	47559.47
ggs	8271	406299.66	774620.75
ipn	4579	224967.68	206195.00
math	3712	671145.37	218070.89
phys	197745	19346378.88	15553409.08
spg	51	2382.60	12832.00
ssb	1410	124178.65	17796.75

Dept	No. of GPU Jobs Submitted	CPU Hrs	GPU Hrs	MEM(GB)
aoes	2282	26133.00	178370.83	565710.00
biol	6	0.10	4.34	600.00
cds	3749	76146.21	848276.45	83665.93
chem	435	26292.79	44722.03	12196.50
ggs	1194	10528.99	45760.75	115584.00
math	345	602.87	4438.51	27218.23
phys	1182	12014.18	137725.83	51606.00
ssb	86	8240.33	98882.87	3857.00

COS Submitted GPU Jobs by Department



COS Submitted GPU Jobs by Department



Instructional Utilization



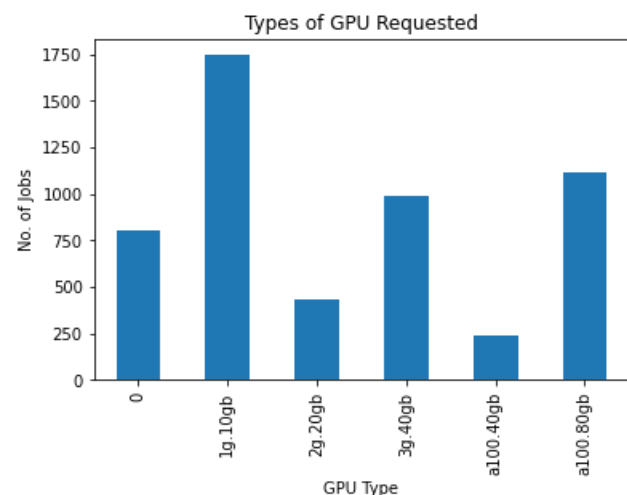
Instructional Accounts: No. of Students and Jobs Submitted

Account	Dept	College	No. of Students	Submitted Jobs
clim680fl23	aoes	COS	4	447
cs678fl23	cs	CEC	21	561
cs700fl23	cs	CEC	1	83
cs701fl23	cs	CEC	1	163
cs747fl23	cs	CEC	2	137
daen690fl23	daen	CEC	18	1190
stat788fl23	stat	CEC	1	32
ait526sp24	ist	CEC	1	2
ait580sp24	ist	CEC	1	2
cs678sp24	cs	CEC	31	1604
cs682sp24	cs	CEC	2	112
cs782sp24	cs	CEC	1	93
csi702sp24	cds	COS	4	502
daen690sp24	daen	CEC	19	348
ece555sp24	ece	CEC	36	2274
stat778sp24	stat	CEC	7	673
Total			150	8223

Instructional GPU Jobs Summary

Account	No. of Students	No. of Jobs Submitted	CPU Hrs	GPU Hrs	MEM(GB)
ait526sp24	1	2	17.40	17.40	80.00
clim680fl23	1	32	2673.52	230.12	2664.00
cs678fl23	20	430	24809.84	4186.69	19987.63
cs678sp24	31	1539	41912.62	6522.29	62004.03
cs682sp24	2	112	11503.06	991.40	8840.00
cs701fl23	1	158	3980.81	156.15	10133.21
cs747fl23	2	133	10146.55	873.72	8568.00
cs782sp24	1	91	2348.27	264.97	2991.00
csi702sp24	2	4	16.30	2.20	200.00
daen690fl23	12	917	54281.38	3725.63	81430.84
daen690sp24	12	241	10995.56	868.24	15777.60
ece555sp24	35	1671	4595.43	2103.85	17452.00

- The largest share of instructional accounts for GPU jobs belongs to the computer science department but the CS department accounts have neither the single highest number of CPU hours nor the single highest number of jobs submitted

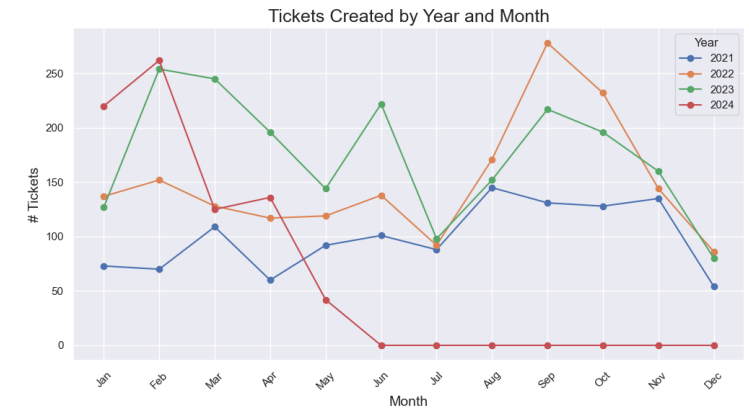
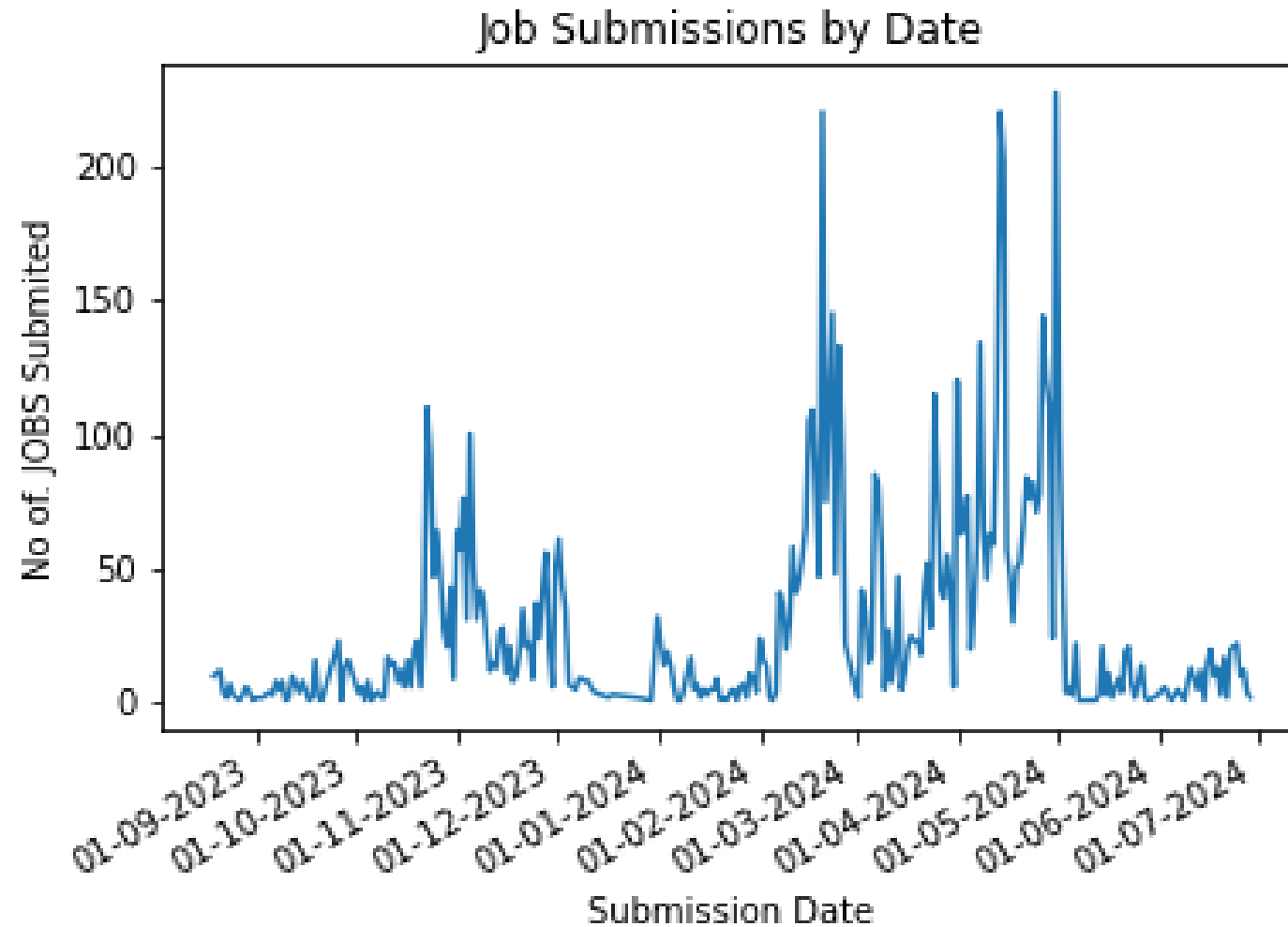


Instructional CPU Jobs Summary

Account	No. of Students	No. of Jobs Submitted	CPU Hrs	MEM(GB)
ait580sp24	1	2	0.40	4.00
clim680fl23	4	415	13005.64	16618.00
cs678fl23	14	131	2460.79	5576.00
cs678sp24	19	65	826.72	1488.00
cs700fl23	1	83	134.96	1382.00
cs701fl23	1	5	7.56	198.00
cs747fl23	2	4	72.05	58.00
cs782sp24	1	2	2.33	24.00
csi702sp24	4	498	36903.16	15392.00
daen690fl23	15	273	18605.54	18006.00
daen690sp24	17	107	817.31	2363.00
ece555sp24	33	603	37960.98	2960.00
stat778sp24	7	673	1185.45	8516.05
stat788fl23	1	32	616.97	890.00

- The Computer Science Department has the largest share of instructional accounts, though the single instructional account with the highest number of CPU jobs belongs to the Stats department
- Despite the large number of CS instructional accounts, they tend to have a small number of students for CPU jobs

Instructional Jobs submission trends



Conclusions

- CEC/CS run the most jobs on the cluster, consistently.
- Most CPU jobs are by the Stats department (CEC).
- Most GPU jobs are submitted by the CS department (CEC).
- For class instruction, most students running jobs are also from CEC
 - CS 678, DAEN 690, ECE 555
 - While CS instructional accounts combined usually have the highest statistics, individually they are often exceeded in number of jobs, students, or amount of memory by other departments